

INTERVAL ESTIMATION AND HYPOTHESIS TESTING IN REGRESSION

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Ordinary Least Squares, OLS, estimation is one of the core elements of regression analysis. Beyond obtaining point estimates of the unknown parameters, in statistics, we are often interested in making inferences about the underlying population relationships. To do so, it is necessary to understand the sampling distributions of the OLS estimators and the hypothesis tests used to assess their statistical significance.

THE DISTRIBUTION OF THE LEAST SQUARES ESTIMATORS

Consider the general form of a multiple regression model, that is, $y_i = \beta_1 + \beta_2 x_{2i} + \beta_3 x_{3i} + \dots + \beta_K x_{Ki} + e_i$. Let's add the 6th assumption that the random errors, e_i have normal distributions, then the dependent variable, y_i , is normally distributed, $y_i \sim N(\beta_1 + \beta_2 x_{2i} + \dots + \beta_K x_{Ki}, \sigma^2) \Leftrightarrow e \sim N(0, \sigma^2)$. Since the least squares estimators are linear functions of dependent variables, it follows that the least squares estimators are also normally distributed, $b_k \sim N(\beta_k, \text{Var}(b_k))$. Therefore, we can form the standard normal variable Z , $z = \frac{b_k - \beta_k}{\sqrt{\text{Var}(b_k)}} \sim N(0, 1)$, for $k = 1, 2, \dots, K$. Replacing the variance of b_k with its estimate, then $t = \frac{b_k - \beta_k}{\sqrt{\text{se}(b_k)}} = \frac{b_k - \beta_k}{\text{se}(b_k)} \sim t_{(n-K)}$, where the number of degrees of freedom is $n - K$.

We can also form a linear combination of the coefficients as $\lambda = c_1 \beta_1 + c_2 \beta_2 + \dots + c_K \beta_K = \sum_{i=1}^K c_k \beta_k$. Thus, we obtain $t = \frac{\hat{\lambda} - \lambda}{\text{se}(\hat{\lambda})} = \frac{\sum c_k b_k - \sum c_k \beta_k}{\text{se}(\sum c_k b_k)} \sim t_{(n-K)}$.

If the errors are not normally distributed, then the least squares errors will not be normal distributed. However, by the CLT, it can be approximately normal in large samples. Thus, it does not stop us from using a t -test, but it does mean to be cautious if the sample size is not large. A test for normally distributed errors, for example the Jarque-Bera test, can be implemented.

INTERVAL ESTIMATION IN REGRESSION

Since we have discussed that the least squares estimators will be a t -distribution, then a $100(1 - \alpha)\%$ C.I. for b_k is $\left[b_k \pm t_{(n-K; \frac{\alpha}{2})} \text{se}(b_k) \right]$. If we want to use interval estimation for a linear combination of coefficients, we define λ in terms of expectation, such that it is a function of $\beta_1, \beta_2, \dots, \beta_K$ for some k . Therefore, a point estimate of λ is $\hat{\lambda}$, i.e., replace β 's with b 's. Thus, a $100(1 - \alpha)\%$ C.I. for $\hat{\lambda}$ is $\left[\hat{\lambda} \pm t_{(n-K; \frac{\alpha}{2})} \text{se}(\hat{\lambda}) \right]$.

What happens if λ is a nonlinear combination of coefficients? Consider $\hat{\lambda} = f(b_i, b_j)$, for $i, j \in \{1, 2, \dots, K\}$ for some nonlinear function f . We can use the delta method to calculate the approximate variance, $\text{Var}(\hat{\lambda}) = \left(\frac{\partial \lambda}{\partial \beta_i} \right)^2 \text{Var}(b_i) + \left(\frac{\partial \lambda}{\partial \beta_j} \right)^2 \text{Var}(b_j) + 2 \left(\frac{\partial \lambda}{\partial \beta_i} \right) \left(\frac{\partial \lambda}{\partial \beta_j} \right) \text{Cov}(b_i, b_j)$. In R, to find the variance-covariance matrix, we can use the function `vcov()` on the model. Thus, a $100(1 - \alpha)\%$ C.I. for $\hat{\lambda}$ is $\left[\hat{\lambda} \pm t_{(n-K; \frac{\alpha}{2})} \text{se}(\hat{\lambda}) \right]$.

HYPOTHESIS TESTING IN REGRESSION

4.1 Testing the Significance of a Single Coefficient

We need to ask whether the data provides any evidence to suggest that y is related to each of the explanatory variables. If a given explanatory variable, x_k , has no bearing on y , then $\beta_k = 0$. Thus, the null hypothesis is $H_0 : \beta_k = 0$ against the alternative hypothesis, $H_1 : \beta_k \neq 0$. The test statistic becomes $t = \frac{b_k}{\text{se}(b_k)} \sim t_{(n-K)}$ since we assume $\beta_k = 0$. We then compare the t critical values with the level of significance, α , from the distribution table to conclude.

4.2 Testing the Significance of a Linear Combination of Coefficients

We set up a hypothesis test for $\lambda = c_1 \beta_1 + c_2 \beta_2 + \dots + c_K \beta_K$ for some k . The test statistic is $t = \frac{\hat{\lambda} - \lambda}{\text{se}(\hat{\lambda})} \sim t_{(n-K)}$, and follow the same method as above.

4.3 Joint Hypothesis Testing

A null hypothesis with multiple conjectures, expressed with more than one equal sign, is called a joint hypothesis. For example, should a group of explanatory variables be included in a particular model? Does the quantity demanded of a product depend on the prices of substitute goods, or only on its own price? Here, we have multiple conjectures, a test of H_0 is a joint test for whether all conjectures hold simultaneously, for example, two conjectures, $\beta_i = 0, \beta_j = 0$ for some $i, j \in \{1, 2, \dots, K\}$. Therefore, a null hypothesis of $H_0 : \beta_i = 0, \beta_j = 0$ is called the restricted model, which is obtained assuming the parameter restrictions in H_0 are true, against $H_1 : \beta_i \neq 0$ or $\beta_j \neq 0$ or both are non-zero, which is called the unrestricted model since the restrictions in the hypothesis have not been imposed on the model.

The F -test for the null hypothesis is based on a comparison of the sums of squared errors, also known as the sum of squared least squares residuals, from the unrestricted model and the restricted model, SSE_U and SSE_R , respectively. What the F -test does is to assess whether this restriction is sufficiently large to be significant. The F -statistic determines what constitutes a large reduction or a small reduction in the sum of squared errors, that is:

$$F = \frac{\frac{(SSE_R - SSE_U)}{J}}{\frac{SSE_U}{(n-K)}} \sim F_{(J, n-K)},$$

where J is the number of restrictions, n is the number of observations, and K is the number of coefficients in the unrestricted model. If the null hypothesis is true, then the F -statistic has what is called an F -distribution with J numerator degrees of freedom and $n - K$ denominator degrees of freedom. If the null hypothesis is not true, then the difference between SSE_R and SSE_U becomes large. The restrictions placed on the model by the null hypothesis significantly reduces the ability of the model to fit the data.

4.4 Testing the Overall Significance of a Model

Consider a general multiple regression model with $K - 1$ explanatory variables and K unknown coefficients. To examine whether we have a viable explanatory model, we set up the null and alternative hypotheses as $H_0 : \beta_2 = 0, \beta_3 = 0, \dots, \beta_K = 0$ and $H_1 : \text{At least one of the } \beta_k \text{ is non-zero for } k = 2, 3, \dots, K$, respectively. Since we are testing whether or not we have a viable explanatory model, this test is also referred to as a test of the overall significance of the regression model. Given that the t -distribution can only be used to test a single null hypothesis, we use the F -test for testing the joint null hypothesis.

The restricted model, assuming the null hypothesis is true, becomes $y_i = \beta_1 + e_i$. The least squares estimator of β_1 in this restricted model is $b_1^* = \frac{\sum y_i}{n} = \bar{y}$. The restricted sum of squared errors is $SSE_R = \sum (y_i - b_1^*)^2 = \sum (y_i - \bar{y})^2 = SST$. Thus, to test the overall significance of a model, but not in general, the F -test statistic can be modified to become:

$$F = \frac{\frac{(SSE_R - SSE_U)}{J}}{\frac{SSE_U}{(n-K)}} = \frac{\frac{(SST - SSE)}{K-1}}{\frac{SSE}{(n-K)}} = \frac{\frac{SSR}{(K-1)}}{\frac{SSE}{(n-K)}} = \frac{MSR}{MSE} \sim F_{(K-1, n-K)},$$

Do note that the squared term of the t -statistic is identical to the F -value, which holds if $J = 1$.

CONCLUSION

The statistical properties of OLS estimators is one of the core elements for inference in regression analysis. Under the assumption that the errors are normally distributed, the estimators are normally distributed too. This allow us to construct confidence intervals, test the significance of individual coefficients and linear combinations of coefficients, and evaluate joint restrictions and the overall significance of a regression model. Even when normality does not hold, large sample data can provide approximate justification for these methods, making them indispensable for empirical analysis.

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